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Event Reconstruction from Forensic Timeline with Formal Event Modeling

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***Abstract*—Digital forensic investigations require structured methods to analyze heterogeneous system logs and reconstruct accurate timelines of user activities. This study presents a lightweight formal model for the reconstruction of digital forensic events using extraction, mapping, and analysis operators. The method transforms heterogeneous web and system logs into unified entities, relations, and a filtered event timeline. The method applies temporal correlation to distinguish meaningful user actions from automated background processes. It reveals clear behavioral phases such as browser activities, software installation, PowerShell execution, and system shutdown. Experiments with multiple forensic case studies show that the model consistently isolates high-relevance events, with low fitness and generalization scores that indicate its deliberate focus on evidence-driven reconstruction.**

***Index Terms*—formalized event model, digital forensics, event reconstruction, ontology, temporal analysis, forensic timeline.**

I. Introduction

Cybercrime continues to increase along with the development of digital technology. This affects various sectors, including individuals, organizations, and governments [1]. According to the Check Point Research 2023 Security Report, global cyberattacks increased by 38% in 2022 compared to 2021, with an average of 1,168 attacks per organization recorded weekly worldwide in that year [2]. The increase in cybercrime has led to a wider range of research on digital forensic analysis.

Digital forensics is the scientific process of identifying and investigating evidence contained within digital devices, with the goal of uncovering relevant facts in an investigation [3]. In addition to identifying evidence, digital forensics also includes reconstruction of digital forensic events to establish the sequence of events based on recovered data. However, this reconstruction process faces challenges, namely the large size and heterogeneity of the data [4]. Large data volumes not only increase investigators' workload, but can also impact every stage of an investigation. This digital evidence can include server logs, file metadata, communication recordings, and even user devices. Data heterogeneity can be divided into three categories: format, temporal, and semantic. Format issues relate to differences in data writing methods between different sources. Temporal issues arise from using sources originating from multiple machines whose time and time zones are not synchronized. Meanwhile, semantic issues are related to differing interpretations of the same event.

In recent years, attention in the field of digital forensics has shifted from approaches based on investigator experience to techniques supported by a strong theoretical foundation. This theory-based approach emphasizes the importance of formal definitions and standards in the digital forensic reconstruction process to preserve the integrity of investigative results [5]. A clear and well-defined investigative model not only helps explain the process used to obtain results, but also strengthens the reliability of the resulting information. Before the development of this theory-based approach, digital forensic analysis often relied on investigator experience and intuition [6].

To address challenges such as large data sizes, investigative complexity, and limitations of the IT infrastructure, Adderley and Peterson [7] proposed an interactive temporal analysis framework that reduces manual event correlation using graph-based visualization and abstraction. Similarly, ontology-driven methods have evolved to integrate semantic reasoning and event correlation in digital forensics [8], [9]. These research efforts collectively emphasize the need for a formal ontology-based modeling framework capable of integrating heterogeneous data, supporting semantic reasoning, and automating event analysis on-scale.

Contribution. In this study, an ontology-based formal model is applied to analyze a digital forensic event reconstruction. This model is designed to address challenges in event reconstruction, such as data heterogeneity and the complexity of entity-to-entity relationship analysis. This approach not only supports the chronological organization of the data, but also supports semantic analysis to identify causal relationships and

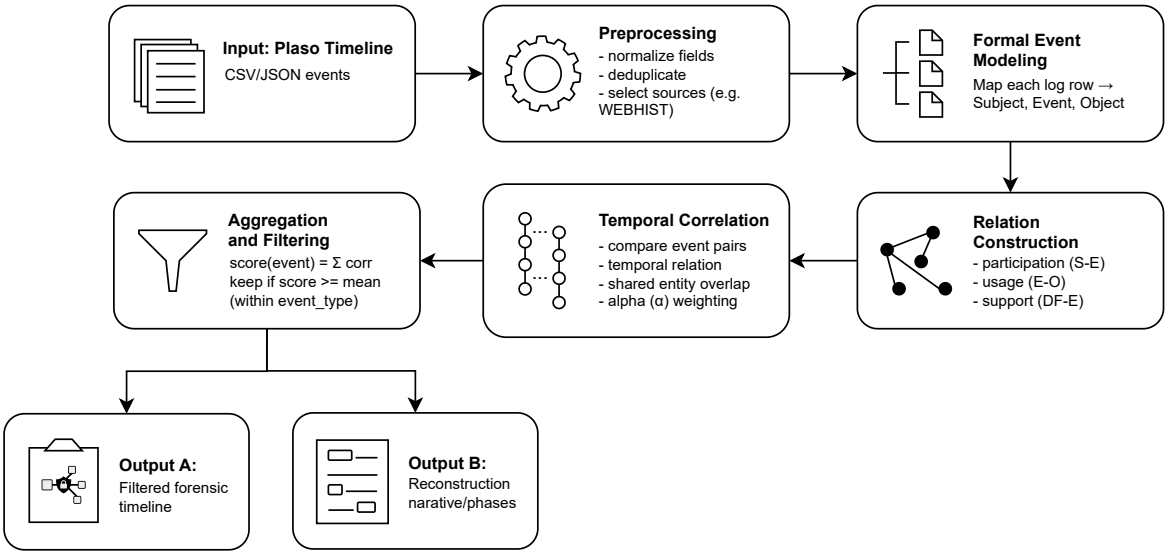

Fig. 1. The proposed method for the forensic event reconstruction from log2timeline plaso CSV data

relevant patterns within the data [6]. This study not only proposes a theoretical approach but also evaluates its practical application in digital forensics.

This paper is organized as follows. Section II reviews related work on event reconstruction and ontology-based modeling approaches. Section III presents the proposed method. Section IV describes the evaluation procedure and reports the results of the reconstructed timeline. Finally, Section V concludes the paper and outlines limitations and future work.

II. Related work

Reconstruction of digital forensic events has been studied for a considerable time. Gladyshev and Patel [10] introduce a finite state machine (FSM)-based reconstruction system and evaluate it using synthetic data. They show that FSMs can work well in relatively simple settings, but the study does not establish whether the sequence produced truly matches the event data. The FORE (Forensics of Rich Events) system developed by Schatz et al. [11] uses the ontology to represent events. However, this method has drawbacks, such as limited semantic coverage and low accuracy. In another work, researchers report similar results with an FSM approach on synthetic datasets, again demonstrating good performance for low-complexity systems [12].

The mCRL2 process algebra is applied to reconstruct events from a FAT file system, and it is shown that the approach can produce a plausible sequence [13]. A more recent method reconstructs events using sufficient and necessary evidence, analyzed using linear temporal logic (LTL) and the check of the NuSMV model [14]. By validating the final state of the system, it improves efficiency over earlier techniques and is reported to be fast and accurate, but it depends on having a complete and correct formal system model. To address time correctness, another study [15] introduces time anchors to assess the accuracy of the system clock during reconstruction; experiments involving Google search and file creation demonstrate its ability to detect time drift.

Regarding timestamp manipulation challenges, Vanini et al. [16] present a qualitative study with ten students tasked with swapping the order of two events on a live Windows 10 system, finding that such tampering is difficult due to tool limitations, the expertise required, and the additional traces created by the manipulation itself. Studies have also demonstrated abstraction-based and symmetry-driven methodologies to improve the understanding of event timelines and relationships [17]. More recent reviews have consolidated terminology, methodologies, and open challenges in timeline-based digital forensic reconstruction [18].

In this study, the approach and methods implemented refer to previous research by Chabot et al. [6] and we apply it to a forensic timeline. The study focused on a formal knowledge representation model, namely the Semantic Analysis of Forensic Cases (SADFC). The method combines digital forensic expertise with software development to create a rich semantic and ontology representation of incident-related events.

III. Proposed method

This work applies a formal ontology-based model to reconstruct digital forensic timelines from the log2timeline Plaso tool [19] as shown in Fig. 1. The method involves collecting relevant digital traces (e.g., system logs or web history) and transforming them into structured entities that represent events, objects, and subjects. We enrich the extracted knowledge to clarify relationships among entities and finally constructing and analyzing the timeline using the formal model and Allen's interval algebra to determine accurate temporal ordering of events. Each step is discussed in the following subsections.

A. Forensic Timeline Collection and Preprocessing

The forensic timeline used in this study consists of 15 CSV files that were collected [20]. We then converted to JSON

TABLE I
RULE-BASED MAPPING OF FOOTPRINT TYPES TO SUBJECT AND OBJECT

Footprint type	Subject	Object
Web Visit	Browser	URL
Process Creation	Executable	Plugin
Program Opened	Program	Plugin
Recent File Access	File	Entry shell details or LNK metadata
Google Search	Browser	Search term
Shutdown Time	Windows	Event ID
Bing Search	Browser	Search term
File Downloaded	User	File name

format by filtering rows based on the source column with a value such as "WEBHIST" (web history). The digital data is collected from various sources relevant to the incident being investigated and consists of digital footprints that provide important information for further analysis.

B. Formal Event Modeling

This stage is the entity creation stage, i.e., events, objects, and subjects, which is the basis for building an ontology for knowledge representation. The ontology ensures that the represented knowledge is consistent, structured, and universally understandable within the context of a specific domain. In this context, the ontology acts as a semantic framework that maps the relationships between various entities involved in a digital incident. By using ontology, investigators can organize digital data in a more structured manner, understand the relationships between events, and generate deeper insights into the incident.

In this work, mapping operators are used to model knowledge representation. There are several procedures in forming the ontology, namely combining all rows of data from each JSON file into the *footprints_df* variable, extracting objects, subjects, and events from digital traces based on their type, filtering duplicate objects and subjects, updating the values of objects and subjects in the *events_df* variable, listing all entities that have the same support relationship, registering support relationship pairs, registering participation relationship pairs, and registering usage relationship pairs.

Consolidating all rows. This step merges all rows from the 15 JSON files into a single data frame (*footprints_df*) to simplify and speed up analysis. The consolidated data frame stores key forensic fields such as IDs, time ranges, evidence source, event details (type, description, or category), plugin and file metadata, correlation attributes (supporting or trigger), and the origin filename. The footprint type is in Table I.

Extracting objects, subjects, and events. This stage extracts objects, subjects, and events from digital traces using rule-based mapping from the type column across eight footprint types (e.g., Web Visit, Program opened, searches, downloads, process creation, shutdown). The process outputs three data frames: *objects_df* (id, details, support), *subjects_df* (id, details, support), and *events_df* (reduced event fields including time plus linked subjects or objects via a shared support identifier). Since each footprint always yields one object, one subject, and one event, no additional knowledge enrichment is needed, and the entity IDs remain continuous.

Filtering duplicate objects and subjects. Because each footprint initially produces one object and one subject (1009 rows each), duplicates appear across the objects and subjects data frames. This step removes duplicates to create *cleaned_objects_df* and *cleaned_subjects_df*, keeping the same columns but renaming the support to *aggregated_support* to store one or more supporting footprint references for each unique object or subject.

Updating object and subject values. After removing duplicate objects and subjects and updating their IDs, this step revises the *events_df* entries so that the objects and subjects fields reference the new unique IDs. So each event links to the correct updated object and subject.

List all entities that have the same support relationship. In this step, we group entities that originate from (or are supported by) the same digital footprint. Let S, O, and E denote the sets of subjects, objects, and events, respectively. We define the set of all entities as:

$$En = S \cup O \cup E. \tag{1}$$

Because a single entity may be supported by one or more footprints (e.g., after deduplication when multiple traces refer to the same object or subject), we represent support as a set-valued mapping:

$$\sup : En \to 2^F, \tag{2}$$

where F is the set of footprints and $\sup(en)$ returns the set of footprints that support entity en. In other words, each entity in the model can be traced back to one or more original digital footprints that justify its existence.

Registering support relationship pairs. This step explicitly records which digital footprints support (i.e., are used to derive or justify) which entities, providing traceability from raw traces to the constructed subject, object, and event entities. Using the set-valued support mapping $\sup(en)$, we define the support relation as a set of ordered pairs:

$$\sigma_{\text{sup}} = \{\, (f, en) \mid en \in En,\ f \in \sup(en) \,\} \ \subseteq\ F \times En, \tag{3}$$

where each pair $(f, en) \in \sigma_{\text{sup}}$ indicates that footprint f supports entity en. After registering these pairs, entities originating from the same footprint can be grouped by f:

$$\text{Group}_{\text{sup}}(f) = \{\, en \in En \mid (f, en) \in \sigma_{\text{sup}} \,\}, \quad \forall f \in F. \tag{4}$$

Thus, $\text{Group}_{\text{sup}}(f)$ returns all entities that share the same supporting footprint, which is later used as a basis for constructing additional relationship pairs (e.g., participation and usage).

Registering participation relationship pairs. This step captures which subjects are involved in each event, creating an explicit subject–event participation record. Because an event may involve one or more subjects, we represent the subjects associated with an event using a set-valued mapping:

$$\text{subj} : E \to 2^S, \tag{5}$$

TABLE II
ALLEN'S TEMPORAL RELATIONSHIP FOR CORRELATION COMPUTATION

Relation	Condition (event e vs. event x)
equals	$e_{end} = x_{start}$
overlaps	$e_{start} < x_{start}$ and $e_{end} > x_{start}$
during	$e_{start} > x_{start}$ and $e_{end} < x_{end}$
finishes	$e_{end} = x_{end}$
before	$x_{start} > e_{end}$

where E is the set of events, S is the set of subjects, and $\mathrm{subj}(e)$ returns the set of subjects participating in event e. The participation relation is registered as a set of ordered pairs:

$$\sigma_{\text{par}} = \{\, (s, e) \mid e \in E,\ s \in \mathrm{subj}(e) \,\} \ \subseteq\ S \times E, \quad (6)$$

where each pair $(s, e) \in \sigma_{\text{par}}$ indicates that subject s participates in event e.

Registering usage relationship pairs. This step captures which objects are used or affected by each event. Because an event may involve one or more objects, we represent the objects associated with an event using a set-valued mapping:

$$\mathrm{obj} : E \to 2^{O}, \quad (7)$$

where E is the set of events, O is the set of objects, and $\mathrm{obj}(e)$ returns the set of objects used by event e. Subsequently, the usage relation is registered as a set of ordered pairs:

$$\sigma_{\text{use}} = \{\, (e, o) \mid e \in E,\ o \in \mathrm{obj}(e) \,\} \ \subseteq\ E \times O, \quad (8)$$

where each $(e, o) \in \sigma_{\text{use}}$ indicates that event e uses object o.

C. Event and Timeline Reconstruction

At this stage, analysis operators are used to identify relationships between events or to highlight relevant information from the timeline. In this study, operators dedicated to identifying correlations between events are introduced as shown in Table II. There are two processes in reconstructing a timeline: creating a *timeline_df* from the *events_df* and calculating the correlations between events. The first step builds *timeline_df* by copying key fields (ID, time, type, subject, object) from *events_df*, then removes duplicate rows where multiple events share the same time, type, subject, and object despite different IDs to preserve accuracy.

The second stage computes correlations between events using the study's analysis operators and Allen's interval algebra to reveal temporal relationships that support deeper analysis. Rule-based correlation is not used because no rules were defined. The parameter α is set to 1 since the data contain only start times (no end times), so start and end are treated as equal; increasing α could weaken the analysis.

D. Relationships Between Events Analysis

Following the timeline reconstruction process, analyzing the relationships between events is a step in understanding the interconnections between the various events. This analysis explores how one event influences or relates to another. This is particularly useful in analyzing patterns or trends emerging from the data, as well as identifying anomalous events.

In the analysis approach, the correlation of each event with other events is calculated. For instance, if there are 593 events, so each event will have 592 correlations with other events. All correlations for each event are summed, resulting in a total correlation value for each event.

After calculating the total correlation, the next step is to divide the events by their respective types. For each event type group, the average of the total correlations obtained is calculated. Events with a total correlation greater than or equal to the average are retained for further analysis, while the remaining events are filtered out. This reduces the number of events to 136. Findings can be drawn on the basis of emerging patterns and the evidence supporting those conclusions.

IV. EXPERIMENTAL RESULTS

A. Results and analysis

The source code for the proposed method is available on GitHub[1]. Following the data processing and formal modeling steps described in the previous sections, the implementation produced several intermediate and final outputs to support verification and analysis. Specifically, the workflow generated a set of Excel files that document the extracted entities, constructed relationships, and correlation-based filtering results:

- Entities: Tables for footprints, objects, subjects, and events, providing a structured overview of the entities.
- Support relations: Support links that trace which footprints to construct particular entities and events.
- Usage relations: Relationships that describe how events are associated with and use specific objects.
- Participation relations: Links that capture the participation of subjects in each event.
- Correlations: Pairwise correlation values computed between events.
- Aggregated correlations: Summarized correlation scores (total correlation per event) derived from pairwise results.
- Timeline and updated timeline: The initial timeline and the filtered timeline after applying correlation-based selection to retain the most relevant events.

The more complete sample event reconstruction is provided in the GitHub repository. As a result of the sample event reconstruction, a sequence of actions revolves around using the Bing browser to access various online resources. The reconstructed activities can be seen in Table III. The subject runs the activity by accessing several JavaScript files hosted on the Bing domain (`r.bing.com`). This pattern is likely to obtain or load additional resources from Bing. In next events, the same URL access pattern continues, with the subject likely inspecting elements of a web page. The subject opens the `verclsid.exe` program, which is typically associated with file security checks to ensure their validity. This may indicate a file verification phase. Next, access switches back to Bing, this time with a URL indicating the presence of interactive

[1]https://github.com/forensic-timeline/reconformal

TABLE III
SAMPLE EVENT RECONSTRUCTION: BING ACTIVITY WITH ADDITIONAL PROGRAM EXECUTIONS

ID	Start	Event	Usage	Participation
1642–1646	2023-12-26 00:35:45	Access Bing JavaScript resource(s) on `r.bing.com` (resource loading)	1102–1106	1455
1656–1658	2023-12-26 00:35:45	Continued Bing URL access; likely inspecting elements of a complex web page	1116–1118	1455
1671	2023-12-26 00:35:50	Program opened: `verclsid.exe` (file verification / security check phase)	1041	1459
1676, 1682	2023-12-26 00:35:50	Return to Bing; URL indicates interactive elements (e.g., data transfer / tracking)	1133, 1134	1455
1727	2023-12-26 00:35:59	Process executed: `OneDriveSetup.exe` (possible reconfiguration/reinstallation)	1011	1460
1736–1739	2023-12-26 00:36:03	Repetitive loading of Bing JavaScript-based resources on `r.bing.com`	1184–1187	1455
1750	2023-12-26 00:36:04	Repetitive loading of Bing JavaScript-based resources on `r.bing.com`	1198	1455
1759	2023-12-26 00:36:04	Program opened: `sc.exe` (Windows service management tool)	1041	1461

elements, such as data transfer or user tracking. Subsequently, the subject executes a process for OneDriveSetup.exe, possibly indicating a reconfiguration of the OneDrive service for file synchronization. The web access activity continues with a similar URL from Bing, which indicates a repetitive pattern of loading JavaScript-based resources. Finally, the `sc.exe` program is opened. This program is a system-level tool that is often used to manage services in Windows, such as starting or stopping system services.

In another sample reconstruction (Table IV), the user searches Bing Search for "mozilla firefox download". This search results in several visits to Mozilla pages, including the download of various related files such as CSS, JS, images, and other assets. Subsequently, the `Firefox Installer.exe` file is successfully downloaded, indicating that the user is preparing to install Firefox. Then, the `crashreporter.exe` program is opened, and recent file accesses related to Mozilla Firefox are visible in the Recent Files directory. The accessed files have certain attributes, such as file size, icon location, and cmd arguments. They indicated that Firefox is running in various modes, including the New Tab Mode, Main Browser Mode, and Private Window Mode.

B. Performance Measurement

We measure reconstruction performance using two quantitative metrics: fitness and generalization. Fitness evaluates how well the model represents relevant data while accounting for event filtering. Higher values mean more relevant events are retained, though lower scores can be acceptable when the model intentionally focuses on only the most important events. Generalization assesses the model's ability to infer plausible unobserved events consistent with existing evidence and patterns, with the aim of logically expanding the reconstruction without over-inference, overfitting, or adding noise.

The evaluation results are presented in Table V. Let I denote the number of plausible inferred events, M the number of missing relevant events, and R the total observed relevant events, The generalization (G) and fitness (F) are defined as $G = \frac{I}{R}$ and $F = 1 - \frac{M}{R}$.

C. Discussion

Forensic timeline data processing for event reconstruction. The timeline data was converted from multi-line CSV to structured JSON to improve organization and enable richer relationship-based analysis. For instance, the dataset was then filtered to keep only "WEBHIST" entries to narrow the focus to web history activity relevant for event reconstruction. As a result, the data became more manageable and supported clearer insights into temporal web-history patterns.

Formal models on the forensic timeline for the reconstruction process. The results show that formal models can support the reconstruction of digital forensic timelines by automating data collection and processing. Using structured elements (e.g., entity tables, relation, and correlation frameworks), the approach systematically identifies and groups relevant events and reduces manual effort that aid interpretation. The relatively low fitness values (0.20-0.36) reflect the model's deliberate filtering strategy rather than a limitation in the reconstruction capability. The correlation-based selection retains only events whose temporal and entity relationships exceed the mean threshold within each event type. In forensic analysis, the vast majority of system log entries represent automated background processes that are irrelevant to the investigation. By filtering these out, the model reduces the event set to a forensically actionable subset.

V. CONCLUSION AND FUTURE WORK

This research has successfully addressed the problem related to the reconstruction of forensic events, the application of formal models, and the measurement of reconstruction performance. The study successfully extracted and processed digital traces from a forensic timeline. These traces were filtered, merged, and converted into structured JSON files. Next, in terms of the formal model application, an ontology-based formal model was applied to the forensic timeline data. This model enabled the definition of entities, identification of relationships among them, and chronological analysis of correlated events to support accurate reconstruction of incidents.

Future work should improve the formal model with more advanced analysis methods, clearer interpretation, prioritize robust handling of corrupted data, further investigate rule-based correlation to deepen event-relationship insights, and develop automated timeline visualization tools. In addition, future work should include a systematic comparative evaluation against existing reconstruction techniques, such as FSM-based and temporal-logic-based methods, using standardized benchmark datasets and metrics.

TABLE IV
SAMPLE EVENT RECONSTRUCTION: BING SEARCH FOR FIREFOX AND SUBSEQUENT DOWNLOAD/EXECUTION-RELATED ACTIVITY

ID	Start	Event	Usage	Participation
1898, 1912, 1915, 1918, 1934	`2023-12-26 00:36:46`	Bing search session for "mozilla firefox download"; visits to Mozilla-related pages and retrieval of page resources	1161, 1314–1316, 1161	1455
1945, 1954–1963	`2023-12-26 00:36:47`	Continued Mozilla page/resource loading following the Bing search results	1323, 1326–1327, 1329	1455
2041	`2023-12-26 00:36:51`	Further navigation/resource retrieval related to Mozilla/Firefox download workflow	1355	1455
2089	`2023-12-26 00:36:56`	Further navigation/resource retrieval related to Mozilla/Firefox download workflow	1370	1455
2203	`2023-12-26 00:37:08`	File downloaded: `Firefox Installer.exe` (download completed; preparation for installation)	1396	1467
2270	`2023-12-26 00:38:50`	Follow-up Firefox-related activity after the installer download (continued browsing/interaction)	1419	1455
2300–2306	`2023-12-26 00:42:54`	Program opened: `crashreporter.exe` and Recent File Access entries related to Mozilla Firefox (e.g., New Tab mode, main browser mode, private window mode)	1041, 1428–1430	1482–1484

TABLE V
PERFORMANCE OF RECONSTRUCTION USING THE DATASET FROM [20]

No.	Scenario / File	Fitness (F)	Gen. (G)
1	All JSON files combined	0.22934	0.22934
2	2-search-firefox.json	0.35889	0.35889
3	3-launch-cmd.json	0.21239	0.21239
4	4-open-mozilla-org.json	0.20339	0.20339
5	5-firefox-installer-download.json	0.20168	0.20168

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